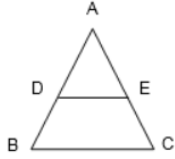


GEOMETRY AND MENSURATION – 1

Directions for question 1: Select the correct alternative from the given choices.



In a triangle ABC, $AB = 12$ cm and $AC = 18$ cm. D and E are points on AB and AC such that DE is parallel to BC. If $AD = 4$ cm, find AE.

- ☐ a) 3 cm
- ☐ b) 4 cm
- ☐ c) 5 cm

☒ d) 6 cm (Your answer - ✓)

Directions for question 2: Type in your answer in the input box provided below the question.

Find the perimeter (in cm) of a rhombus whose diagonals measure 48 cm and 14 cm.

Solution:

The length of the diagonals AC and BD of the rhombus ABCD measure 48 cm and 14 cm. If the diagonals intersect at M, then $AM = 24$ cm, and $BM = 7$ cm, $AB^2 = AM^2 + BM^2 = (24^2 + 7^2) \text{ cm}^2$. $\therefore AB = 25$ cm. The perimeter is $4(25)$ cm, i.e. 100 cm.

Ans : (100)

Directions for questions 3 to 5: Select the correct alternative from the given choices.

What is the radius (in cm) of a wheel which covers a distance of 51.7 m in 47 revolutions?

- ☐ a) 35

☐ b) 17.5 (Correct Answer - ✓)

- ☐ c) 3.5
- ☐ d) 1.75

Solution:

The wheel covers a distance of 51.7 m in 47 revolutions. If r is the radius of the wheel,
 $47(2\pi r) = 51.7\text{m}$

$$\Rightarrow r = \frac{7(51.7)}{22(47)(2)}\text{m} = \frac{7(5170)}{22(47)(2)}\text{cm} = 17.5\text{ cm}$$

$\therefore$ The length of the radius is 17.5 cm.

Choice (B)

Directions for questions 3 to 5: Select the correct alternative from the given choices.

The supplement of an angle is five times the angle. The angle measures ____.

- ☐ a) 30°
- ☐ b) 35°
- ☐ c) 40°
- ☐ d) 45°

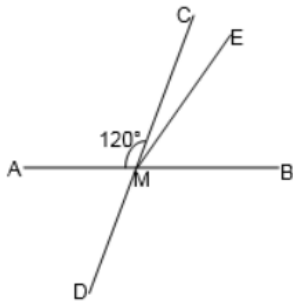
Solution:

The supplement of an angle (say θ) is 5 times the angle.

$$\therefore 180^\circ - \theta = 5\theta \text{ or } \theta = 30^\circ.$$

Choice (A)

Directions for questions 3 to 5: Select the correct alternative from the given choices.



In the above figure

$\angle$

$\angle BMD = 8$

$\angle$

CME and AB intersects CD at M. Find

$\angle$

CME.

☐ a) 30°

☐ b) 35°

☐ c) 20°

☒ d) 15° (Your answer - ✓)

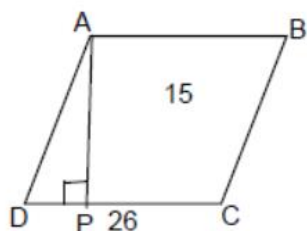
Directions for questions 6 and 7: Type in your answer in the input box provided below the question.

The base of a parallelogram is 26 cm. The length of the perpendicular drawn from the base to its parallel side is 15 cm. What is the area of the parallelogram? (in sq.cm)

Your Answer: Unattempted

Correct Answer: 390 ✓

Solution:



In parallelogram ABCD, $CD = 26$ cm and $AP = 15$ cm

$$\therefore \text{Area of ABCD} = 26(15) \text{ cm}^2 = 390 \text{ cm}^2$$

Ans : (390)

Directions for questions 6 and 7: Type in your answer in the input box provided below the question.

What is the length (in cm) of the inradius of a triangle of sides 3 cm, 4 cm and 5 cm?

Your Answer: Unattempted

Correct Answer: 1 ✓

Directions for questions 8 and 9: Select the correct alternative from the given choices.

Find the area (in sq.cm) of a circle inscribed in a square of area 288 sq. cm.

- ☐ a) 288π
- ☐ b) 144π
- ☐ c) 36π

☒ d) 72π (Correct Answer - ✓)

Solution:

The area of the square is 288 cm^2 . $\therefore$ Its side is $12\sqrt{2}$ cm. The radius of the inscribed circle is $6\sqrt{2}$ cm. The area of the inscribed circle is $72\pi \text{ cm}^2$. Choice (D)

Directions for questions 8 and 9: Select the correct alternative from the given choices.

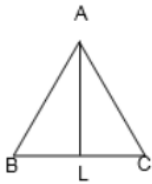
Find the ratio of the volumes of two cuboids whose lengths, breadths and heights are in the ratio 7 : 5, 5 : 6 and 3 : 14 respectively.

- ☐ a) 4 : 3
- ☐ b) 4 : 1
- ☐ c) 3 : 4
- ☐ d) 1 : 4 (Correct Answer - ✓)

Solution:

The lengths, breadths and the heights of the two cuboids are in the ratio 7 : 5, 5 : 6 and 3 : 14 respectively. $\therefore$ The volumes are in the ratio $7(5)(3) : 5(6)(14) = 1 : 4$.
Choice (D)

Directions for question 10: Type in your answer in the input box provided below the question.



AL is the bisector of

$\angle$

BAC. If $BL = 4$ cm, $LC = 3$ cm and $AC = 7.5$ cm, find AB (in cm).

Your Answer: Unattempted

Correct Answer: 10 ✓

Solution:



The figure AL bisects $\angle BAC$. $BL = 4$ cm, $LC = 3$ cm, $AC = 7.5$ cm. From the angle bisector theorem, $\frac{AB}{AC} = \frac{BL}{LC}$

$$\therefore AB = AC \left(\frac{BL}{LC} \right) = 7.5 \left(\frac{4}{3} \right) \text{ cm} = 10 \text{ cm}$$

Ans : (10)

Directions for question 11: Select the correct alternative from the given choices.

The ratio of the angles of two sectors cut from a circle is 2 : 3. Find the ratio of the lengths of the respective arcs.

☐ a) 3 : 2

☐ b) 4 : 9

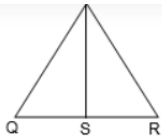
☒ c) 2 : 3 (Correct Answer - ✓)

☐ d) 9 : 4

Solution:

The length of an arc of a circle is proportional to the radius of the circle of which it is a part and the angle subtended at the center. Here, the two arcs are from the same circle. Therefore, their ratio is the ratio of the central angles, i.e. 2 : 3.

Choice (C)



In the above figure, the line PS bisects

$\angle$

QPR, and

$\angle$

PSQ =

$\angle$

PSR.

$\angle$

SPR =

$\angle$

PRS + 10° . Find

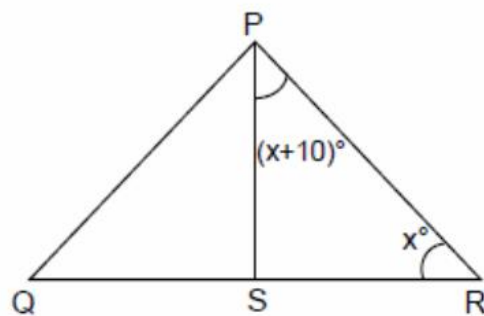
$\angle$

QPR (in degrees).

Your Answer: Unattempted

Correct Answer: 100 ✓

Solution:



PS bisects $\angle$ QPR.

$\angle$ PSQ = $\angle$ PSR

$\therefore$ Each is a right angle.

$\angle$ SPR = $\angle$ PRS + 10°

As $x + (x + 10) = 90^\circ$, $x = 40^\circ$

$\angle$ QPR = $2(x + 10)^\circ = 100^\circ$

Ans : (100)

Directions for question 13: Select the correct alternative from the given choices.

Find the ratio of the volumes of a hemisphere and a cylinder of the same base area, if the height of the cylinder is eight times the radius of the cylinder.

☒ a) 1 : 12 (Correct Answer - ✓)

☐ b) 1 : 6

☐ c) 1 : 3

☐ d) 1 : 2

Solution:

The hemisphere and the cylinder have the same base area. The height of the cylinder is 8 times the radius of the cylinder.

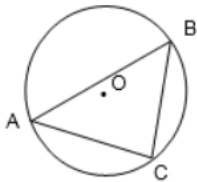
$$\text{Volume of the hemisphere} = \frac{2}{3} \pi r^3$$

$$\text{Volume of the cylinder} = \pi r^2 (8r) = 8\pi r^3$$

$$\text{Ratio} = \frac{2}{3} : 8 = 1 : 12$$

Choice (A)

Directions for question 14: Type in your answer in the input box provided below the question.



In the figure shown, O is the centre of the circle,

$\angle$

$\text{AOC} = 160^\circ$. If $AB = BC$, then find

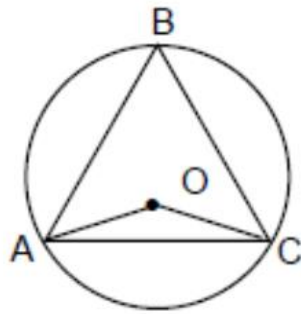
$\angle$

BAC (in degrees).

Your Answer: Unattempted

Correct Answer: 50 ✓

Solution:



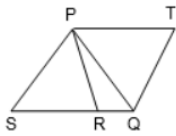
$$\angle AOC = 160^\circ \therefore \angle ABC = 80^\circ$$

$$AB = BC \therefore \angle BAC = \angle BCA$$

$$\text{As } \angle BAC + \angle BCA = 100^\circ, \text{ each has to be } 50^\circ.$$

Ans : (50)

Directions for question 15: Select the correct alternative from the given choices.



In the figure above,

$\angle$

$\angle PSR = 60^\circ$ and

$\angle$

$\angle PRQ = 120^\circ$. Find

$\angle$

$\angle PQR$ given

$\angle$

$\angle RPS = 2$

$\angle$

$\angle RPQ$.

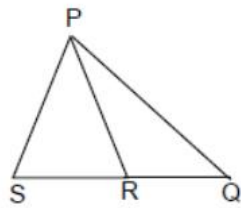
☐ a) 60°

☐ b) 50°

☐ c) 40°

☐ d) 30° (Correct Answer - ✓)

Solution:



Note that T plays no role in either the data or the question.

$\angle PSR = 60^\circ$, $\angle PRQ = 120^\circ$,

$\therefore \angle SPR = \angle PRS = 60^\circ$

$\angle RPS = 2 \angle RPQ \therefore \angle RPQ = 30^\circ$ and hence $\angle Q = 30^\circ$.

Choice (D)